4. **Configure Baud Rate**: The default baud rate is 9600 bps. Ensure your microcontroller's UART settings match this value.

Example: Using GT-U7 with Arduino UNO

Below is an example of how to use the GT-U7 module with an Arduino UNO to read GPS data:

```
#include <SoftwareSerial.h>

// Define RX and TX pins for SoftwareSerial
SoftwareSerial gpsSerial(4, 3); // RX = Pin 4, TX = Pin 3

void setup() {
  Serial.begin(9600);          // Initialize Serial Monitor at 9600 bps
  gpsSerial.begin(9600);       // Initialize GPS module at 9600 bps
  Serial.println("GT-U7 GPS Module Test");
}

void loop() {
  // Check if data is available from the GPS module
  while (gpsSerial.available()) {
    char c = gpsSerial.read(); // Read one character from GPS module
    Serial.print(c);           // Print the character to Serial Monitor
  }
}
```

Best Practices

- **Antenna Placement**: Place the antenna in an open area with minimal obstructions for better signal reception.
- **Power Supply**: Use a stable power source to avoid fluctuations that may affect GPS performance.
- **UART Settings**: Ensure the baud rate and UART settings match between the GT-U7 and your microcontroller.
- **Signal Acquisition**: Allow the module a few seconds to acquire satellite signals, especially during a cold start.

Troubleshooting and FAQs

Common Issues and Solutions

1. **No GPS Data Output**
   - **Cause**: Incorrect wiring or baud rate mismatch.
   - **Solution**: Double-check the connections and ensure the baud rate is set to 9600 bps.
2. **Poor Signal Reception**
   - **Cause**: Antenna is obstructed or placed indoors.